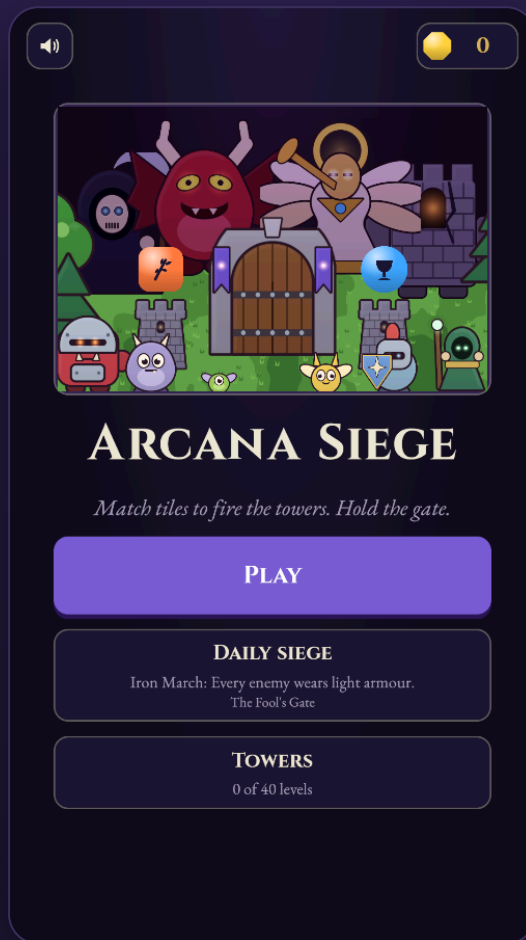


GAME DESIGN DOCUMENT · V1.0

ARCANA SIEGE

A match-3 tower defense. Match tiles to fire your towers; hold the gate against the Arcana.



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Game design, systems, balancing, UX, art and audio direction

September 2026 · Unity 6 · portrait mobile and web

THE GAME IN ONE SCREEN

Top half: a road, four towers and a gate. Bottom half: a match-3 board. Every tile colour belongs to a tower, so every match is an order.

THE PITCH

Enemies march down the road towards the gate. The player never places a tower; they **fire** them. The colour of a match picks the tower, its size picks the strength, and cascades multiply it. The road runs on the player's moves: each move lets the enemies walk 1.6 seconds closer, and while the player thinks they only creep.

PILLARS

- **Every match is a decision.** Read the road, then the board. The right colour at the right moment beats a big match of the wrong one.
- **Choosing beats speed.** Every move costs the same stretch of road, so thinking is cheap and mashing gains nothing (measured: a first-time player wins 50–76% at any pace).
- **A run in five minutes,** ten waves and a boss, always "one more try" close.
- **An ad that explains itself** in two seconds: one big combo wipes a wave off the road.



A siege in the Greenwood

AUDIENCE AND MARKET

Hybrid-casual: match-3 players who enjoy a little pressure, and tower defense players who want a quicker, tactile core. Portrait, one-handed, sessions of 5 to 15 minutes. Match-3 and mobile TD both earn well; a readable blend of the two is rare.

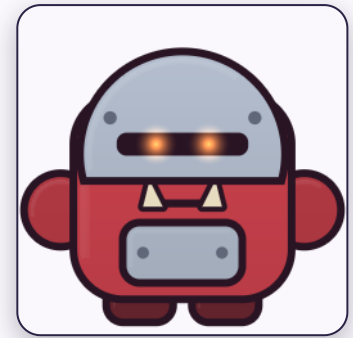
REFERENCES

GAME	WHAT IT DOES WELL	WHAT ARCANA SIEGE TAKES
Royal Match	Fast, clear, generous match-3; bright pieces on calm backgrounds	The board's readability and juice; a saturation hierarchy
Kingdom Rush	Readable tower defense; cel-shaded, framed maps; a theme per region	The art direction, framed realms, a track per realm
Empires & Puzzles	Matched tiles fly to the battlefield; hero mana and ultimates	Motes from board to tower; the Moon charges an ultimate
Rush Royale, Raid Rush	Run-based power growth; towers that look stronger as they level	The 1-of-3 draft; visible tower ranks

ONE MOVE, ABOUT TWO SECONDS

1. **Read the road:** what is coming, how close, how armoured?
2. **Pick a match:** the colour decides which tower fires, the size how hard.
3. **Watch it land:** motes fly from the board to the tower (0.34 s), the tower fires, enemies fall or keep walking.

Each cascade after the first multiplies tower output by another 25%, so a lucky or planned chain is a big moment on both halves of the screen. A match of four hits a third harder than a match of three, because output is per tile.



An armoured Brute on the road

THE ROAD RUNS ON MOVES

Each move buys the enemies **1.6 s of walking**; between moves they creep at 25% (half that while a finger is on the board). Owed walking plays out faster, so mashing only moves the road faster. **Calm mode** stops the road between moves. To help the read, the tiles the road needs pulse with their tower's halo, a pressed tile lights its tower, and a hint appears after 3 s.

SPECIAL TILES BECOME TOWER MOVES

- **Line clearer** (match 4 in a line): its tower fires at *every* enemy on the road, 4 tiles' worth.
- **Wheel** (L or T shape): a quake that stuns every enemy for 1.5 s and hits it for 20.
- **Star** (match 5): starfall, 45 to every enemy.
- **Arcana** (swap two specials): fills the moon and casts the ultimate at once.

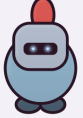
Special moves and the ultimate pierce armour and shields, so the board's big plays answer the road's hardest enemies.

TILES AND TOWERS

TILE	TOWER	EFFECT (PER TILE CLEARED)	ROLE
 Wands	Fire	6 damage to the front enemy and everything within 2.5; burns 1.5/s for 3 s	Groups
 Cups	Tide	2 damage to the whole road; slows 8% (max 60%) for 3 s	Buys time
 Swords	Blade	14 damage to the enemy closest to the gate; ignores armour	Bosses, Brutes
 Pentacles	Ward	Repairs the gate by 7.5	Survival
 Moon	Hero	10 mana; at 100 the moon button calls the ultimate (150 to every enemy)	Emergency
 Sun	Gold	1 gold, spent on re-dealing the draft; banked for permanent upgrades	Economy

EACH ENEMY ASKS A DIFFERENT QUESTION

Base stats (health is multiplied by each siege's scale, and by 1.25 since the playtest pass). Every trait is answered by a different tower, so the road tells the player which colour to look for.

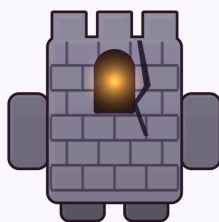
	ENEMY	HEALTH	SPEED	GATE DMG	GOLD	TRAIT	DESIGN INTENT
	Grunt	108	1.1	30	1	—	The baseline.
	Runner	58	2.0	18	1	—	Fast and fragile: punishes ignoring the road. Answer: Tide.
	Brute	432	0.65	72	3	60% armour	Fire and Tide glance off. Answer: the Blade, specials.
	Shieldbearer	90	1.0	30	2	Shield	The first direct hit does nothing. Answer: splash, many small hits.
	Acolyte	84	0.9	24	3	Heals	Every 2.5 s mends 12% of max health to all within 3. Answer: strike it first.
	Broodmother	300	0.7	48	3	Splits	Bursts into 3 Imps where it falls. Answer: kill it early, keep the Tide ready.
	Imp	26	2.3	10	0	—	A Broodmother's young.

CHAMPIONS AND THE ARCANA

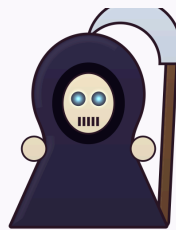
Every siege ends with a boss wave. The last siege of each realm brings one of the **Arcana** (60% armour, very slow, breaks the gate outright if it arrives). The other eight bring a **champion**: a common enemy grown huge and crowned, quicker and lighter-armoured (30%), just as deadly at the gate. The boss track takes over the music the moment either walks on.



The Devil
Greenwood · 3,500



The Tower
Sunsorched · 5,600



Death
Marsh · 6,300



Judgement
Summit · 8,000

Armour and shields are mirror images. Armour rewards one big Blade strike and punishes spraying fire; a shield absorbs the first direct hit whatever its size, so it rewards many small hits and punishes the lone big strike. Together they stop any single colour from being "the right answer".

TEN WAVES, A DRAFT BETWEEN EACH

A siege is ten waves; the tenth brings the champion or boss. After each cleared wave the clock stops for the **draft**: pick 1 of 3 cards; it lasts for the rest of the siege. The board stays playable during the short break that follows.

- **Bad-luck protection:** three offers in a row without a card for your weakest tower guarantee one next.
- **Re-deal** for gold: 10, then +5 each time in the same draft. Sun tiles pay for it.
- Cards stack up to their copy count, so a run can lean hard into one tower.

Worth is the change in win rate, in points, when the first-time bot (2.2 s a move, 63% with a normal draft) holds the card from wave 1 (500 sieges each). Every card helps and none is a must-pick. *The Sun's Tithe pays gold, whose value arrives with the meta.



The draft after wave 1

CARD	SUIT	EFFECT	COPIES	WORTH
Ace of Wands	Fire	Fire hits 40% harder.	3	+5
Wildfire	Fire	Burns last twice as long and burn 50% hotter.	2	+9
Spreading Flame	Fire	Fire reaches 60% further along the road.	2	+3
Ace of Cups	Tide	Slows last twice as long, and the tide hits 50% harder.	2	+17
Undertow	Tide	The tide hits three times harder.	2	+16
Deep Water	Tide	Slows are twice as strong and can reach 85%.	1	+18
Ace of Swords	Blade	The Blade hits 40% harder.	3	+10
Twin Edges	Blade	Each Blade strike also hits the next enemy for 90%.	1	+9
Executioner	Blade	The Blade does 70% more damage to Brutes and the boss.	2	+16
Ace of Pentacles	Ward	Repairs are twice as strong.	3	+2
Bulwark	Ward	The gate gains 30 health, and repairs by 30 now.	3	+7
Stone Skin	Ward	Enemies that reach the gate do 25% less damage.	2	+6
Rising Moon	Moon	Moon tiles give 40% more mana.	2	+10
Eclipse	Moon	The hero's ultimate hits 60% harder.	2	+13
The Sun's Tithe	Sun	Every kill pays 1 more gold.	2	0*
The Wheel Turns	Sun	Cascades build power twice as fast.	2	+13

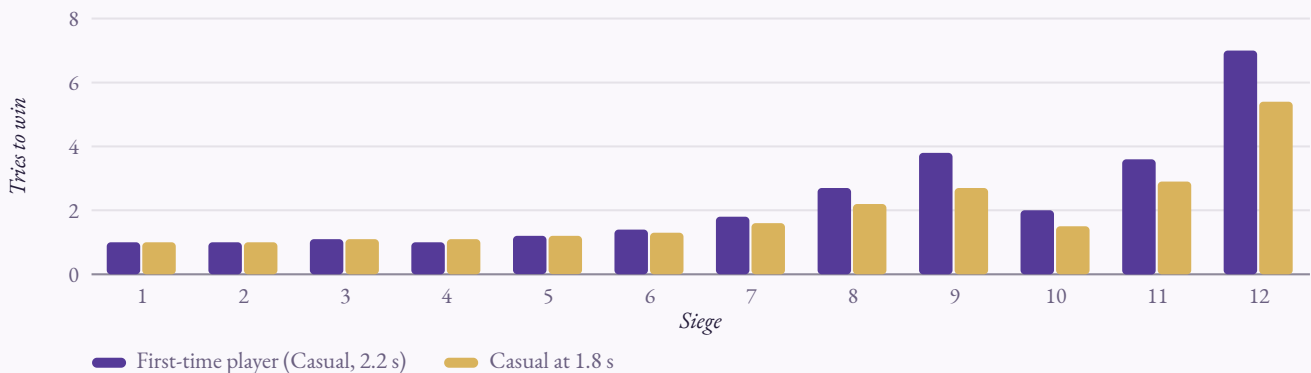
FOUR REALMS, TWELVE SIEGES

The Fool's journey through four realms. Each realm adds one enemy and ends with one of the Arcana.

#	SIEGE	REALM	NEW IDEA	ENDS WITH	HEALTH	TRIES*
1	The Fool's Gate	 Greenwood	grunts and runners only	Old Gnasher	×0.85	1
2	The Magician's Ford	 Greenwood	armoured Brutes	The Ford Warlord	×1.05	1
3	The Devil's Thicket	 Greenwood	everything so far	the Devil	×1.2	1.1
4	The Chariot's Dunes	 Suncorched	Shieldbearers	The Shield Captain	×1.35	1
5	The Hermit's Well	 Suncorched	shields in front of Brutes	The Sand Colossus	×1.5	1.2
6	The Tower's Road	 Suncorched	the realm's finale	the Tower	×1.6	1.4
7	The Hanged Mire	 Marsh	Acolytes (healers)	The High Acolyte	×1.75	1.8
8	The Star's Fen	 Marsh	healers in crowds and behind shields	The Bog Brute	×1.8	2.7
9	The Moon's Keep	 Marsh	everything, denser	Death	×1.85	3.8
10	The Empress's Pass	 Summit	Broodmothers (split into Imps)	The Brood Queen	×2.05	2
11	The World's Edge	 Summit	every enemy at once	The Frost Warlord	×2.1	3.6
12	Judgement Peak	 Summit	the finale	Judgement	×2.15	7

*Average runs until the siege first falls, from the career bot (100 careers each): it starts with nothing, plays the furthest open siege, banks every run's gold and buys the cheapest tower level after each run.

A SAWTOOTH, ON PURPOSE



A champion siege after a realm's finale is a breather (fresh upgrades, a new enemy to learn); each finale is a wall. A first-time player finishes the whole map 95% of the time within 20 tries a siege, in about 28 runs (a little over two hours); a practised one in 16. Judgement at 7 tries is the number to watch in a playtest.

WHY YOU COME BACK

PERMANENT TOWER LEVELS

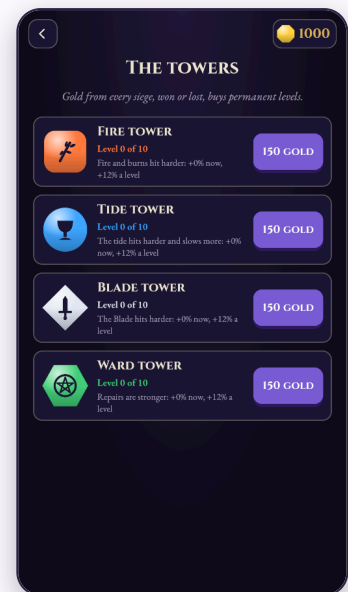
Gold from every siege, won or lost, is banked and buys permanent levels on the Towers screen: +12% output per level, ten levels per tower, $150 \times 1.3^{\text{level}}$ gold each (about 6,400 to max one tower, 25,600 for all four). Losing still moves you forward. On the road, towers show their rank: a pennant at level 3, a crown at 6, a standing aura at 10.

GOLD

- **Sources:** kills (1 to 3, champions and bosses 25), Sun tiles, a win bonus from 80 (siege 1) to 500 (Judgement).
- **Sinks:** draft re-deals inside a run; tower levels between runs.
- A typical siege pays 300 to 550 gold; the first level is affordable after the first siege.

STARS

1 for a win, 2 if the gate stood above half, 3 if it was barely touched. Stars unlock the next siege and give a completionist goal (36 on the map).



The Towers screen

LEVEL	1	2	3	4	5	6	7	8	9	10
Cost	150	195	254	330	428	557	724	941	1,224	1,591
Output	+12%	+24%	+36%	+48%	+60%	+72%	+84%	+96%	+108%	+120%

DAILY SIEGE

One siege from the first two realms, with a modifier (Blood Moon, Gold Rush, Iron March, Swift Feet, Blessed Blade, Glass Gate), seeded by the date so everyone plays the same board and draft. Scored on kills, waves and gate: a reason to open the app every day.

MONETISATION PLAN

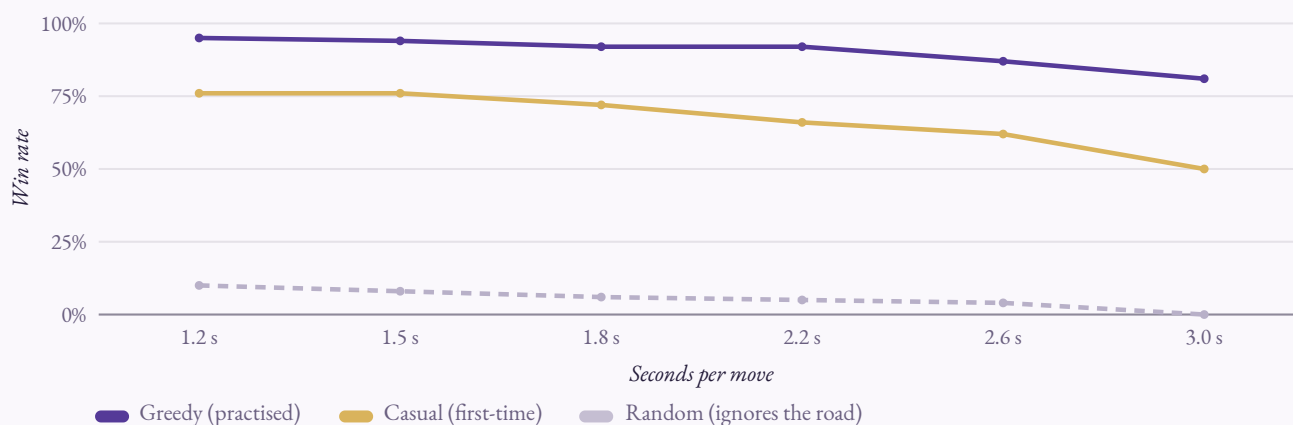
All through `IStoreService` and `IAdService` (placeholders in this build). **Rewarded:** a second chance (the gate rebuilt to 40%, once per siege).

Interstitial: never before the fourth run, never after a daily, every other siege end. **IAP (planned):** remove ads, a starter gold pack, cosmetic tower skins. No energy system.

TUNED BY BOTS PLAYING THE REAL RULES

The rules live in pure C# (`ArcanaSiege.Core` , no Unity), so a console runner compiles the exact game code and plays thousands of sieges in parallel (about 25 seconds for the full report). Three bots stand in for three kinds of player; **pace** is seconds per move, thinking plus animation. In Calm mode (no creep) Casual wins 89%, Greedy 99%, Random 20%.

PLAYER	PACE	TARGET	BEFORE PLAYTEST	NOW
First-time (Casual)	2.2 s	55–65%	63%	66%
Slow thinker (Casual)	3.0 s	40%+	10%	50%
Practised (Greedy)	1.8 s	90%+	98%	92%
Practised but slow (Greedy)	2.6 s	70–85%	68%	87%
Ignores the road (Random)	1.8 s	≤ 25%	23%	6%
Mashes (Random)	1.2 s	lowest	87%	10%



WHAT THE DATA CHANGED

Speed beat judgment. After the draft and specials landed, fast random swiping won 95%. Brutes and bosses now wear armour that turns aside 60% of Fire and Tide; the Blade, specials and the ultimate pierce it. Random play at a practised pace fell from 50% to 22%.

The draft is the run. With no draft the first-time bot wins 2%; with it, 63%. Every card was measured alone: the weakest buffed, the strongest trimmed, until none is dead and none is a must-pick.

A playtester: too much at once. A professional at a mobile studio found reading the road, the towers and the board while the enemies kept walking too demanding, and the data agreed: thinking was punished (a slow first-time bot won 10%). Slowing the road after each move was tried first and rejected (mashers won 99%). Now each move buys a fixed walk, enemies creep between moves, are 25% tougher, and walk slower in the first two realms. Mashing fell from 87% to 10%; slow thinkers rose to 50%.

Still to check with people: whether the pulse of walking after a move reads as natural, and whether the glow and the hint help the choice or make it for the player.

CEL-SHADED STICKERS, FOUR REALMS

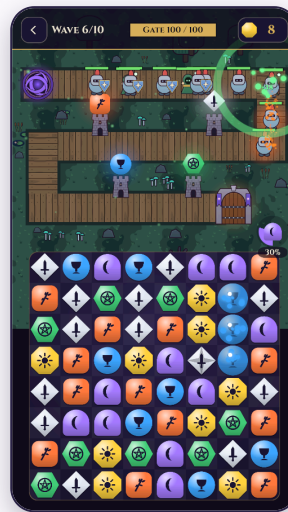
Every character, boss, tower, prop and texture is drawn in code from distance-field shapes, so the art is original, consistent and regenerated in seconds.



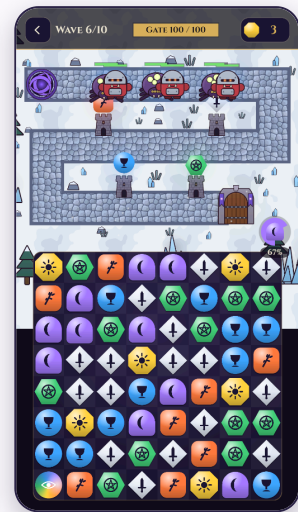
The Greenwood



The Sunscorched Road



The Moonlit Marsh



The Frozen Summit

STYLE

- **Cel shading** as in Kingdom Rush: a base tone, a crisp band of cool shadow on the edges facing away from the light, a crisp glint on the edges facing it. No airbrushed gradients.
- **Warm plum outlines**, never pure black.
- **Saturation hierarchy**: tiles and towers are the most saturated things on screen, enemies next, scenery muted.
- **Silhouette first**: each enemy reads by shape before colour (horns, wings, helmet, shield, hood, egg sac).

REALMS

- **Greenwood**: dirt path, oaks, flowers, pollen.
- **Sunscorched Road**: sandstone, cacti, obelisks, dust.
- **Moonlit Marsh**: boardwalk, reeds, glowing mushrooms, fireflies.
- **Frozen Summit**: icy flagstones, snowy pines, falling snow.

Big props frame each map; small props keep off the road; a dark HUD band with a gold edge keeps text readable on sand and snow.

EVERY HIT SHOULD LAND

MUSIC

Orchestral, one track per place, as tower defense games give each region its own theme: a calm harp and recorder theme on the menus; a heroic, a driving, an urgent and a militant battle track for the four realms; an epic loop that takes over when a boss or champion walks on. Tracks cross-fade, never cut. (CC0 recordings from OpenGameArt, credited in the project.)

EFFECTS

A recorded impact leads where the real thing reads best (steel for the Blade, a rammed door for the gate, plate for a shield, coins for gold), picked from 3 to 5 takes with pitch jitter, and the game's own synthesized sound sits under it. Magic stays synthesized. Match chimes climb a pentatonic scale with the cascade.

MIX

Music at 0.3 against effects up to 0.9, for phone speakers; music streamed and Vorbis-compressed.

JUICE

- **Motes** fly from the cleared tiles to their tower, so the player sees the order being carried out.
- **Hit-stop:** a 0.06–0.12 s freeze on the ultimate, quakes and the death of a Brute or boss.
- **Callouts:** "Cascade ×4" grows warmer and bigger with the chain.
- Hit flashes (a white silhouette of each creature), squash on impact, screen shake on big moments.
- **Glancing blows** on armour show as small grey numbers, teaching the player to switch to the Blade.
- Healing shows as a green ring and "+N"; a broken shield says "blocked".

READABILITY

Tips sit on a dark plate that wraps and shrinks to fit any phone; floating numbers carry a heavy outline and shadow and are kept inside the road, clear of the HUD.

TEACH BY THE ROAD, NOT BY TEXT

FIRST SESSION

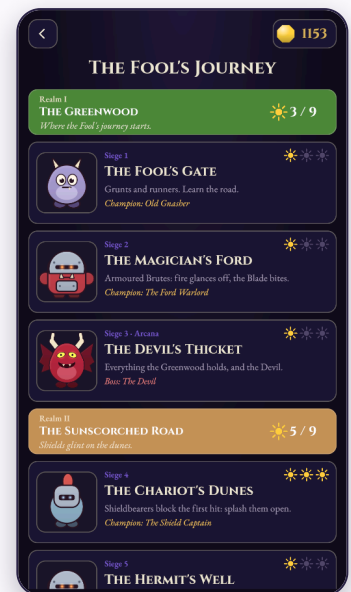
1. **Home:** key art of the four Arcana behind the gate; one big Play button; the daily siege and Towers below.
2. **The map** opens scrolled to the furthest open siege. Each card shows the foe at its end, in shadow until unlocked.
3. **Siege 1** has only grunts and runners: the player learns that colours fire towers, and that runners need watching.
4. **Siege 2** brings the first armoured Brute behind a screen of grunts, with one tip: "Armoured: use the Blade".
5. Each realm's new enemy arrives alone first, with one tip; its answer is a tower the player already knows.

SCREENS

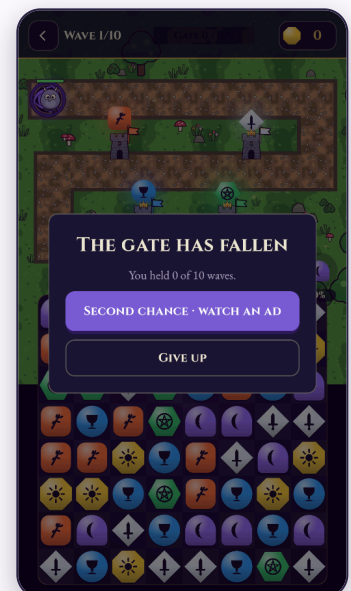
Home, the map, the Towers screen, the siege, the draft panel and the result panel (with the second chance). The game screen is fixed: HUD band, road, the moon button in the free corner by the gate (near the thumb), board below.

ACCESSIBILITY

- Every tile has a colour, a shape *and* a glyph, so no information depends on colour alone.
- No time limit on the draft; the clock stops while you choose.
- Portrait, one-handed; the UI respects the safe area.



The map



The second chance

RULES IN PURE C#, TESTED AND SIMULATED

ARCHITECTURE

- `ArcanaMatch.Core` : the match-3 rules (board, matching, specials, cascades), reused from Arcana Match.
- `ArcanaSiege.Core` : the siege (enemies, towers, waves, draft, campaign, rewards, daily). No UnityEngine, fixed-step, deterministic from a seed.
- `ArcanaSiege.Game` : MonoBehaviours that only display, animate and take input.
- The scene, all art and all synthesized sounds are generated from code by editor menus, so the project rebuilds from source.

QUALITY

- **132 EditMode tests** cover every rule: armour, shields, healing, splitting, the draft's protection, rewards, unlocks, saves.
- The balance runner compiles the same Core code in .NET and plays whole careers.

SERVICES

- Save: a JSON profile with repair for damaged or old saves.
- `IStoreService` , `IAdService` : placeholders now, an SDK later without touching game code.
- `IAalytics` : ten events (funnel, difficulty, draft, economy, ads) logged locally as JSON lines.

BUILDS

- Web (WebGL, gzip, mobile page template, 27 MB zip) for itch.io.
- Android: IL2CPP ARM64, API 26+, portrait.

ROADMAP

- **Done:** the first outside playtest, which led to the move-driven road, Calm mode and the reading help.
- **Next:** a playtest with 10 people to check the bots and the new pace, then a CPI test (see the ad and KPI plan).
- **Content:** a fifth realm per update; tower skins; a weekly event siege.
- **Depth:** a second road fork for placement decisions, if playtests ask for more strategy.